Reply to Michał Bogdziewicz et al.: Research on the summer solstice needs falsifiable predictions

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In [1] we showed that summer solstice appears as an optimal time to transition between major physiological growth and reproductive phases. Given a trade-off between predicting total and remaining thermal heat, summer solstice was optimal on average for Europe and North America, but with local variation. Bogdziewicz et al. interpret our results as invalidating the potential role of solstice in mast seeding [2], but we suggest they have mis-interpreted our work. We explicitly considered whether solstice may control the timing of mast seeding, and our results do not invalidate this hypothesis. In [1] we do, however, consider the potential role of other drivers and raise questions about how to differentiate between them given strong covariation.

Bogdziewicz et al. argue that summer solstice is a better cue for masting than temperature-related cues because it allows for greater synchrony across individuals [2], whereas we highlight spatial variation in a trade-off based on temperature accumulation. We do not disagree that we show important variability across space. We question, however, their conclusions because: (i) in many masting trees, the strength of synchrony among individuals decays with distance, inconsistent with cuing directly to summer solstice [3], and (ii) the strong covariation between summer solstice and balance in annual heat accumulation makes it difficult to reject either as the primary cue—a major message of our paper. Showing a masting pattern consistent with cuing to solstice is not the same as demonstrating trees cue to solstice.

In [1] we discuss the challenge in differentiating between cues. Current work on whether the summer solstice acts as "*celestial starting gun*" has focused on a statistical correlation between an annual metric of seed production and daily temperature to show a shifting correlation may occur near the summer solstice on average, but this timing similarly varies across space ([4], Fig. 1 in [2]). These results echo ours [1] and thus we are confused as to where Bogdziewicz et al. see a discrepancy between these findings and ours. Identifying what level of variation invalidates the hypothesis seems an important question, especially as research on the role of the summer solstice for both growth [5] and masting show results with the most likely date before or after the solstice by one or more weeks.

Understanding what scale of variation to expect on average seems equally important as developing testable predictions for variation across space. While we agree that many economy of scale hypotheses for masting require some level of synchrony to lead to increased fitness, most would not predict such large-scale synchrony [6] and may thus predict cues other than solstice. Even in the case of extremely widespread synchrony, whether the solstice alone would be an ideal cue for masting is not clear from current work.

Developing falsifiable predictions to test the role of the solstice—and potentially other cues—in driving masting will be important for fundamental science and forecasting. We hope our work, which built from eco-evolutionary and climatological perspectives [1], can help, especially as bounded systems (e.g., the growing season, [7, 8]) and natural climatological trends may lead to interesting patterns with multiple competing explanations and different predictions [9, 10].

References

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